

Samuel Kaz

Glen Rock, NJ — samuel.kaz1@gmail.com — 201-364-4242

PROFESSIONAL EXPERIENCE:

- Design Engineer** | Mainstream Fluid & Air, Berkeley Heights, NJ Jun. 25-Present
- Engineered hundreds of custom fan arrays to retrofit AHUs, optimizing reliability and design output quality.
 - Performed airflow, pressure, and spatial analysis to ensure effectiveness, increasing efficiency by 30%.
 - Generated assembly drawings and unit specifications using Draftsight to streamline RFQ approval.
- Systems Integration Lab Research Assistant and TA** | Stevens, Hoboken, NJ May. 24-May. 25
- Developed and refined an ARUCO robotic boat used to autonomously locate and retrieve objects.
 - Implemented an MPU accelerometer integrated with a PID controller using Arduino for automatic maneuvering.
 - Mentored 60+ students through project development, providing baseline code and robot prototypes for support.
- Fiber Engineering Intern** | PPL Corporation, Allentown, PA May. 23-Aug. 23
- Developed a resource website on fiber optic and substation integration, achieving an 80% adoption rate.
 - Quickly learned and utilized company-specific software to complete Splice plans for fiber infrastructure.
- CAD Engineering Intern** | H2 Energy Now Jun. 22-Aug. 22
- Utilized CAD software to blueprint a 12 component assembly of a patented Hydrogen Separation Chamber.
 - Presented research and proposed design options regarding the best suited materials and fabrication methods.

EDUCATION:

- Stevens Institute of Technology** | Hoboken, NJ
- Master of Engineering in Mechanical Engineering, Concentration in Medical Devices* Graduated May. 25
- GPA:** 3.9/4.00 | **Coursework:** Medical Device Design/Manufacturing, Electron Microscopy, Mechatronics
- Bachelor of Engineering in Mechanical Engineering, Minor in Electrical Engineering* Graduated May. 24
- Coursework:** Fluid Dynamics, Microprocessor Systems, Control Systems, Thermal Engineering, Electronic Circuits
- Honors:** Degree with Honor, Dean's List, 2nd place in Gallois Autonomous Robot Competition

PROJECTS:

- Medical Device Design and Manufacturing Projects** Oct. 24-Apr. 25
- Conducted detailed DFM and DFA assessments via disassembly of medical devices for technical analysis, centering design recommendations on risk management, design controls, sterility, and biocompatibility.
 - Analyzed medical device design and manufacturing processes in accordance with ISO 13485:2016 and FDA 21 CFR Part 820, authoring product development plans and a regulatory compliance report for product packaging.
 - Evaluated theoretical Process Validation frameworks and applied FMEA to mitigate device failure modes.
- Earthworm Soft Body Robot (CAPSTONE):** Sep. 23-May. 24
- Led a team to engineer a medical prototype capable of traveling within the brain to perform thrombectomy, reducing required procedural staff by 50% and lowering treatment costs for hospitals.
 - Conducted rigorous research and technical analysis in Nitinol actuation, manufacturing, and biocompatibility.
 - Followed industry product development phases, producing pitch-ready deliverables for investors.
- Automatic Pet Feeder:** Mar. 22-May. 22
- Project coordinator and CAD modeler for a cost-effective automatic food, water and toy dispensing robot.
 - Programmed microcontrollers to regulate various actuating systems based on environmental changes.

SKILLS:

Technical Skills: ISO 13485:2016 Documentation, Project Management, CAD/CAE, Robotics

Software: SolidWorks, Draftsight, MATLAB, CREO, Arena, Arduino, Minitab, Visual Studio, MS Office Suite

Programming: Arduino, MATLAB, C++

EXTRACURRICULARS:

Robotics, Chess, Varsity Tennis, Skiing, Ping Pong, 3D Modeling, Ultimate Frisbee